



DEPARTMENT OF
COMPUTER SCIENCE

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BSc in Computer Science and Engineering

ATTRIBUTE GRAMMARS IN OCAMLFLAT/OFLAT

MASTER IN COMPUTER SCIENCE AND ENGINEERING
SPECIALIZATION IN SPECIALIZATION NAME

NOVA University Lisbon
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[1] The L^AT_EX Project team. *L^AT_EX – A document preparation system*. 1985. URL: <https://www.latex-project.org>.

[2] J. M. Lourenço. *The NOVAthesis L^AT_EX Template User's Manual*. NOVA University Lisbon. 2021. URL: <https://github.com/joaomlourengo/novathesis/raw/main/template.pdf>.

Dedicatory lorem ipsum.

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” *“This is the way.”*
— **Mandalorian**

ABSTRACT

Formal Languages and Automata Theory (FLAT) is rigorous and often perceived as abstract, which complicates early learning. This dissertation extends the **OCamlFLAT** library with support for *attribute grammars*, including synthesized and inherited attributes, evaluation strategies, and static checks to detect common specification errors. The contribution focuses on a reliable, executable core and clear interfaces.

The original plan included enhancements to the **OFLAT** web application and a pedagogical evaluation. In the final scope, the work is limited to the extension of the **OCamlFLAT** library with the necessary structures needed for later integration. The graphical components and the empirical study of pedagogical impact are *out of scope* and are outlined as future work.

The dissertation reviews the relevant background on attribute grammars within the theory of computation, surveys related educational tools, and details the architecture and implementation of the OCamlFLAT extensions. By stabilising the core and defining integration points, the work provides a basis for subsequent visualisation and classroom use.

Keywords: Formal Languages · Automata Theory · OCaml · OFLAT · OCamlFLAT · Functional Programming · Formal Grammars

RESUMO

A Teoria de Linguagens Formais e de Autômatos (FLAT) é rigorosa e frequentemente percebida como abstrata, o que dificulta a aprendizagem inicial. Esta dissertação estende a biblioteca **OCamlFLAT** com suporte para *gramáticas de atributos*, incluindo atributos sintetizados e herdados, estratégias de avaliação e verificações estáticas para detectar erros de especificação comuns. A contribuição centra-se num núcleo executável fiável e em interfaces claras.

O plano original incluía melhorias na aplicação web **OFLAT** e uma avaliação pedagógica. No âmbito final, o trabalho limita-se a estender a biblioteca **OCamlFLAT** com as estruturas necessárias para integração futura. As componentes gráficas e o estudo empírico do impacto pedagógico ficam *fora do âmbito* e são delineadas como trabalho futuro.

A dissertação revê a fundamentação relevante sobre gramáticas de atributos no contexto da teoria da computação, realiza um levantamento das ferramentas educativas relacionadas e descreve a arquitetura e a implementação das extensões ao **OCamlFLAT**. Ao estabilizar o núcleo e definir pontos de integração, o trabalho fornece uma base para posterior *visualização* e utilização em *contexto letivo*.

Palavras-chave: Teoria de Linguagens Formais · Autômatos · OCaml · OFLAT · OCamlFLAT · Programação Funcional · Gramáticas Formais

DISCLOSURE OF USAGE OF GENERATIVE ARTIFICIAL INTELLIGENCE

The author, Pedro Emanuel Inácio Bailão, acknowledges the use of the following generative artificial intelligence tools in the development of this thesis. These technologies were utilized under strict supervision, and all AI-generated outputs were critically reviewed and verified for accuracy. The author accepts full responsibility for the validity and originality of the content, confirming that the use of these tools aligns with all institutional policies and standards of academic integrity.¹

Generative AI tools used: ChatGPT-5.1.

Activities supported by Generative Artificial Intelligence Tools:²

Conceptualization

- | | | |
|--|--|--|
| ☐ Idea generation | ☐ Formulating research questions and hypotheses | ☐ Hypothesis viability assessment |
| ☐ Defining the research objective | ☐ Feasibility assessment and risk evaluation | ☐ Simulated debate and argument testing |

Literature Review

- | | | |
|--|--|--|
| ☐ Search and Discovery | ☐ Concept Mapping and Systematization | ☐ Gap Identification and Novelty Evaluation |
| ☒ Literature summarization and synthesis | ☐ Market/patent landscape analysis | ☐ Cross-lingual literature comprehension |

Methodology

- | | | |
|---|--|--|
| ☐ Experimental Design Optimization | ☐ Development of experimental or research protocols | ☐ Methodological Instrument Selection |
|---|--|--|

¹This declaration was generated using pdfL^AT_EX [2] and the L^AT_EX package aidisclose [1].

²This list extends the GAIDeT taxonomy [3].

Software Development and Automation

- | | | |
|--|--|--|
| ☐ Code Generation | ☒ Debugging and Repair | ☐ Algorithm design |
| ☒ Refactoring and Optimization | ☒ Process automation | ☐ Code documentation and comment generation |

Data Management

- | | | |
|---|--|--|
| ☐ Data collection | ☐ Data analysis | ☐ Synthetic data generation |
| ☐ Validation | ☐ Quantitative Plotting and Charting | ☐ De-identification and anonymization support |
| ☐ Data cleaning | ☐ Reproducibility and rerun checks | ☐ Audio-to-text transcription and diarization |
| ☐ Data curation and organization | ☐ Data labeling and annotation assistance | |

Visuals and Multimedia

- | | | |
|---|--|--|
| ☐ Synthetic Asset Generation | ☐ Image Enhancement and Editing | ☐ Diagrammatic and Schematic Design |
|---|--|--|

Writing and Editing

- | | | |
|--|---|---|
| ☐ Drafting Text | ☐ Formulation of conclusions | ☒ Citation formatting and bibliography management |
| ☒ Polishing and Editing | ☐ Tone Adjustment | ☐ Press releases and outreach materials |
| ☐ Abstract and Executive Summary Generation | ☒ Translation | ☐ Title Generation |

Ethics Review

- | | | |
|--|---|--|
| ☐ Bias analysis and discrimination assessment | ☐ Ethical risk analysis | ☐ Data confidentiality monitoring |
| | ☐ Monitoring compliance with ethical standards | |

Quality Assurance

- | | | |
|--|--|--|
| ☐ Simulated Peer Review | ☐ Identification of limitations | ☐ Publication support |
| ☐ Consistency checking against field trends | ☒ Publication strategy and journal selection | |

Additional comment #1: The AI was used primarily for refining the code in Section 3, but the algorithm logic was derived manually.

Additional comment #2: This is a second additional comment!

Additional comment: This is a third additional comment! It is starred and, thus, does not get a number. Also, this comment is longer and, hopefully, will span to the next line!

Additional comment #3: This is the third non-starred additional comment!

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ARTIFICIAL INTELLIGENCE DISCLOSURE STATEMENT

The `novathesis` template supports the rendering of a *Artificial Intelligence Disclosure Statement*.

The rendering is controlled by the `\ntsetup{print/aidisclosure=OPTION}` option in file `0-Config/1_novathesis.tex`.

Possible values for `OPTION` are:

false	Will omit the AI Disclosure Statement.
filename.tex	Will render the given file as the AI Disclosure Statement.
aidisclose	Will render the <i>standard</i> AI Disclosure Statement with the help of the <code>aidisclose</code> package.

1.1 Deactivating the AI Disclosure Statement

In file `0-Config/1_novathesis.tex` add the option `\ntsetup{print/aidisclosure=false}`.

1.2 Using Your Own AI Disclosure Statement

In file `0-Config/1_novathesis.tex` add the option `\ntsetup{print/aidisclosure=filename.tex}`.

1.3 Using the `novathesis` Standard AI Disclosure Statement

The `novathesis` template uses the `aidisclose` package [1], which extends the *GAIDeT (Generative AI Delegation Taxonomy)* [3] to automate Generative AI disclosure statements and checklists.

The `aidisclose` configuration file can be found at `0-Config/7_aidisclose.tex`. Although you can edit/change this file, it is strongly recommended that you use the package's companion web site at aidisclose.org, illustrated in Figure 1.1.

Fill the Author name and AI tools used, optionally add comments, leave *Configuration & Rendering* as is, and then select the appropriate checkboxes in *Taxonomy of Delegated Tasks*. Finally, press the green button at the bottom, and replace the `novathesisaidisclose` configuration file (`0-Config/7_aidisclose.tex`) with the contents given in the web site.

Generative AI Disclosure Generator

Generate compliant LaTeX code for the `aidisclose` package (v1.11.0).

General Information

Authors *

e.g. Jane Doe, John Smith

Full names of authors accepting responsibility.

AI Tools Used

e.g. ChatGPT-4, GitHub Copilot, Grammarly

Comma separated list of tools.

Comments & Details

+ Add Specific Comment

Configuration & Rendering

☒ Auto-generate BibTeX file (autobib)

☒ Suppress footnote citation (nocite)

☐ No Title (\AIDrenderDeclarations)

Columns

Language

3

Default

Custom Section Title (Optional)

Default: Generative AI Disclosure

Short Title (TOC)

\section, \chapter...

LaTeX Label

e.g. sec:aid_declaration

Taxonomy of Delegated Tasks

Select specific tasks delegated to Artificial Intelligence.

CONCEPTUALIZATION

☒ Idea generation

☐ Defining objectives

☐ Research questions

☒ Feasibility assessment

☐ Hypothesis viability

☐ Simulated debate

LITERATURE REVIEW

☐ Search & Discovery

☐ Summarization

☐ Concept Mapping

☐ Patterns & Themes

☐ Identifying Gaps

☐ Translation

METHODOLOGY

☐ Research Design

☐ Prototyping

☒ Validation

SOFTWARE & AUTOMATION

☐ Code Generation

☒ Optimization

☒ Debugging

☒ Scripts

☐ Algorithms

☐ Documentation

DATA MANAGEMENT

☐ Collection/Scraping

☐ Validation/Quality

☐ Cleaning

☐ Curation

☐ Analysis

☐ Visualization

☐ Reporting Results

☐ Labeling

☐ Synthetic Data

☐ Anonymization

☐ Translation

VISUALS

☐ Image Generation

☐ Editing

☒ Charts/Diagrams

WRITING & EDITING

☒ Drafting

☒ Polishing

☐ Summarizing

☐ Connecting Ideas

☒ Tone Adjustment

☐ Translation

☐ References

☐ Proofreading

☐ Title Suggestions

ETHICS

☐ Bias Mitigation

☐ Risk Assessment

☐ Compliance

☐ Conflict of Interest

CRITIQUE & FEEDBACK

☐ Q&A / Querying

☐ Troubleshooting

☐ Identifying Limits

☐ Future Work Recs

☐ Response to Reviewers

Generate Full Document

Generate Snippet Only

Generate `NOVAthesis`
`7_aidisclose.tex`

Help keep this tool alive

Donate PayPal

Figure 1.1: The aidisclose.org web site

1.3.1 About the AI Tags

1.4 Detailed Taxonomy Descriptions

This appendix provides a detailed breakdown of the `aidisclose` taxonomy. For each category, we define the **Objective** (the high-level goal of using AI in this phase) and the **Scope** (what is generally included or excluded).

1. Conceptualization

Objective: To use AI as a thought partner for brainstorming, refining the research direction, and establishing the theoretical foundation before empirical work begins.

Scope: Includes ideation, hypothesis formation, and feasibility checks. Excludes the actual execution of experiments or data collection.

Keys:

Idea generation: Using AI to brainstorm new research topics, interdisciplinary connections, or novel angles on existing problems.

Objective refinement: Refining vague goals into concrete, actionable research objectives.

Research questions: Drafting and iterating on specific research questions (RQs) to ensure they are clear and answerable.

Feasibility check: Assessing whether the proposed study is viable regarding resources, time, and data availability.

Preliminary research: Conducting quick background checks or “pre-studies” to see if the idea has already been solved.

Simulation/Scenarios: Using AI to conceptualize theoretical models, simulate persona responses, or design hypothetical scenarios.

2. Literature Review

Objective: To accelerate the discovery, synthesis, and organization of existing knowledge.

Scope: Includes searching, summarizing, and translating papers. Excludes the final critical argumentation (which remains the author’s responsibility).

Keys:

Search & Discovery: Using AI tools (e.g., semantic search) to find relevant papers that keyword searches might miss.

Summarization: Generating summaries or abstracts of long papers to quickly assess relevance.

Mapping: Visualizing connections, citation networks, or thematic clusters in the literature.

Pattern recognition: Identifying trends or recurring themes across a large corpus of text.

Gap identification: Using AI to suggest areas where current research is lacking or contradictory.

Translation: Translating foreign-language literature to make it accessible for the review.

3. Methodology

Objective: To assist in the structural design of the research study.

Scope: Includes experimental design and instrument creation. Excludes the physical conduct of experiments.

Keys:

Experimental design: Designing the logic, control groups, and variables of the study.

Prototyping: Creating early drafts of survey instruments, interview guides, or experimental apparatus designs.

Method selection: Suggesting appropriate statistical methods or qualitative frameworks for the data.

4. Software Development and Automation

Objective: To facilitate the creation, optimization, and maintenance of code used in the research.

Scope: Includes coding, debugging, and documentation.

Keys:

Code generation: Generating boilerplate code, scripts, or functions from natural language descriptions.

Optimization: Refactoring code for better performance or readability.

Debugging: Identifying syntax errors or logical bugs in scripts.

Automation: Writing scripts to automate file management, backups, or batch processing.

Algorithm design: Assisting in the logic and mathematical formulation of algorithms.

Documentation: Generating docstrings, comments, and README files for research software.

5. Data Management

Objective: To handle the data lifecycle from collection to reporting.

Scope: Includes cleaning, analysis, and synthetic generation. Excludes the fabrication of results (which is ethical misconduct, distinct from declared synthetic data).

Keys:

Collection: Writing scrapers or using AI agents to gather public data.

Validation: Checking data for consistency, outliers, or errors.

Cleaning: Automating the formatting, parsing, and repair of messy datasets.

Curation: Organizing and categorizing large datasets.

Analysis: Suggesting or performing statistical tests and interpreting raw outputs.

Visualization: Generating code for plots, graphs, and data dashboards.

Reporting: Summarizing data findings in textual or tabular format.

Labeling: Using LLMs to annotate or classify text/image datasets (zero-shot/few-shot labeling).

Synthesis: Generating synthetic datasets to preserve privacy or augment small samples.

Anonymization: Detecting and removing Personally Identifiable Information (PII).

Translation: Translating textual data (e.g., survey responses) into the analysis language.

6. Visuals and Multimedia

Objective: To create or enhance non-data visual elements.

Scope: Includes illustrative diagrams and image editing. Excludes scientific data plots (covered in Data Management).

Keys:

Generation: Creating conceptual images, illustrations, or diagrams from scratch.

Editing: Enhancing, cropping, or modifying existing images (e.g., removing background).

Charts/Infographics: Creating flowcharts, process diagrams, or high-level infographics.

7. Writing and Editing

Objective: To assist in the textual articulation of the research.

Scope: Includes drafting, polishing, and translation. Note: Authors remain accountable for accuracy.

Keys:

Drafting: Generating initial text for sections based on bullet points or notes.

Polishing: Correcting grammar, spelling, and punctuation.

Summarizing: Creating the abstract or plain-language summary.

Conclusions: Synthesizing the discussion into a final concluding statement.

Tone adjustment: Rewriting text to be more formal, concise, or accessible.

Translation: Translating the manuscript from the author's native language to the publication language.

References: Formatting citations and bibliography styles.

Presentation: Drafting slide decks or conference poster text.

Title generation: Brainstorming catchy and accurate titles for the work.

8. Ethics Review

Objective: To act as a check on the ethical integrity of the work.

Scope: Includes bias detection and risk assessment.

Keys:

Bias detection: Scanning text or study designs for potential cultural or gender bias.

Risk assessment: Identifying potential dual-use concerns or societal risks.

Compliance: Checking against specific ethical guidelines or checklists.

Confidentiality: Ensuring no private data is inadvertently leaked in the text.

9. Quality Assurance (Supervisor Role)

Objective: To use AI as a critical reviewer or “devil’s advocate.”

Scope: Includes simulated peer review and limitation checking.

Keys:

Simulated Peer Review: Using AI as a reviewer to identify unclear sections or potential critiques.

Field Trends & Consistency: Checking the manuscript against current research trends and ensuring internal consistency.

Limitations: Identifying weaknesses or limitations in the study that the authors missed.

Publication strategy: Suggesting suitable journals or conferences and helping with the submission strategy.

Publication support: Assisting in drafting cover letters, responses to reviewers, or other publication-related materials.

BIBLIOGRAPHY

- [1] . J. M. Lourenço. *The aidisclose package: Generative AI disclosure checklist and statements*. Version 1.11.0. 2025. URL: <https://ctan.org/pkg/aidisclose/aidisclose-doc.pdf> (cit. on pp. xv, 1).
- [2] The L^AT_EX Project team. *L^AT_EX – A document preparation system*. 1985. URL: <https://www.latex-project.org> (cit. on p. xv).
- [3] Yana Suchikova and Natalia Tsybuliak and Jaime A. Teixeira da Silva and Serhii Nazarovets. “GAIDeT (Generative AI Delegation Taxonomy): A Taxonomy for Humans to Delegate Tasks to Generative Artificial Intelligence in Scientific Research and Publishing”. In: *Accountability in Research* (2025), pp. 1–27. DOI: [10.1080/08989621.2025.2544331](https://doi.org/10.1080/08989621.2025.2544331) (cit. on pp. xv, 1).



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Attribute Grammars in OCamlFLAT/OFLAT

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